1 Write down the exact value of cos (0°)

1

(Total for Question 1 is 1 mark)

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2 Write down the exact value of sin (60)

$\frac{\sqrt{3}}{2}$

(Total for Question 2 is 1 mark)

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3 3 tins of beans and 4 tins of tomatoes costs £2.44
5 tins of beans costs £1.60

Work out how much one tin of tomatoes costs.

$1.60 \div 5 = 0.32$

$3 \times 0.32 = 0.96$

$2.44 - 0.96 = 1.48$

$1.48 \div 4 = 0.37$

£0.37

(Total for Question 3 is 2 marks)

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4 In a company the ratio of men to women is 3:4
40% of the women are under the age of 30.

What fraction of all the people in the company are women under the age of 30?

$\frac{4}{7}$ women

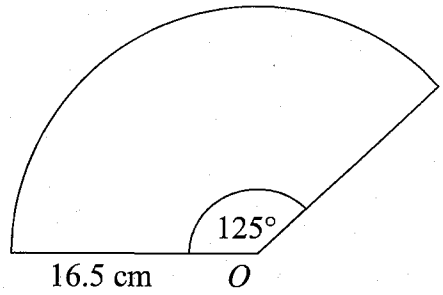
40% of $\frac{4}{7}$

$\frac{2}{5} \times \frac{4}{7} = \frac{8}{35}$

$\frac{8}{35}$

(Total for Question 4 is 3 marks)

5 The diagram shows a sector, centre O.
The radius of the circle is 16.5 cm.
The angle of the sector is 125°.



Calculate the area of the sector.
Give your answer correct to 3 significant figures.

$$\frac{125}{360} \times \pi (16.5)^2 = 297 \text{ cm}^2$$

297 cm^2

(Total for Question 5 is 3 marks)

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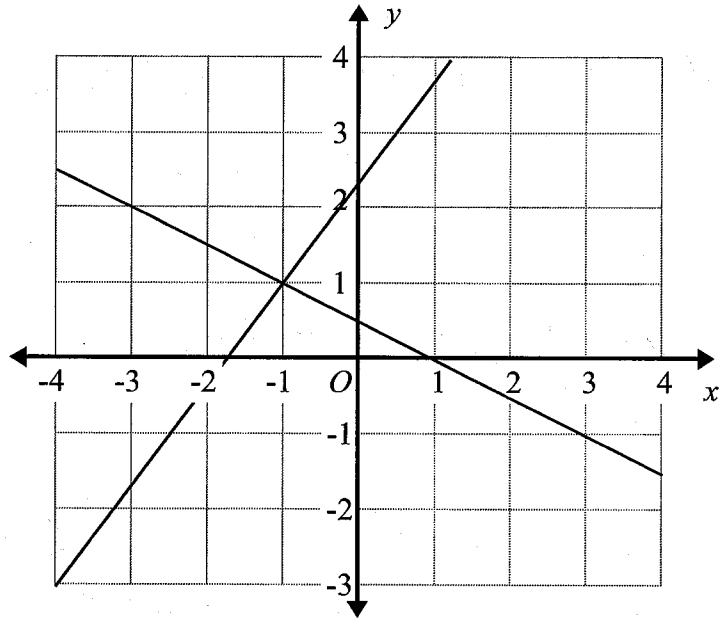
6 The graphs of the straight lines with equations $3y = 4x + 7$ and $x + 2y = 1$ have been drawn on the grid.



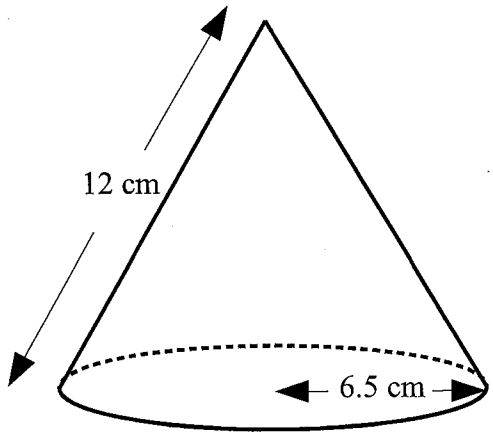
Use the graphs to solve the simultaneous equations

$$3y = 4x + 7$$
$$x + 2y = 1$$

$x = -1$ $y = 1$

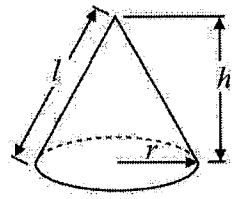
(Total for Question 6 is 2 marks)

7 The diagram shows a solid cone.



Volume of cone $= \frac{1}{3}\pi r^2 h$

Curved surface area of cone $= \pi r l$



The slanted height of the cone is 12 cm.
The base of the cone has a radius of 6.5 cm.

Work out the total surface area of the cone.
Give your correct to 3 significant figures.

Area of circle $= \pi(6.5)^2$

$= 132.73 \text{ cm}^2$

Curved s.a $= \pi(6.5)(12)$

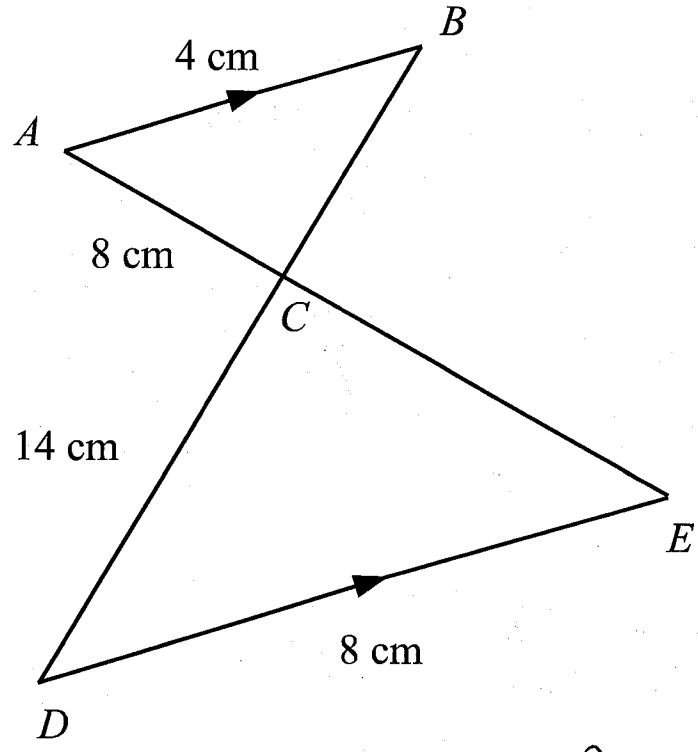
$= 245.04 \text{ cm}^2$

$132.73 + 245.04 = 378 \text{ cm}^2$

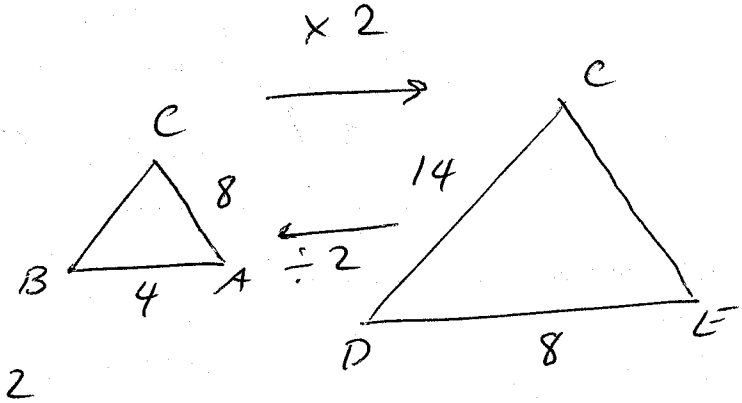
378 cm^2

(Total for Question 7 is 4 marks)

8



AB is parallel to *DE*.
ACE and *BCD* are straight lines.
AB = 4 cm,
AC = 8 cm,
CD = 14 cm,
DE = 8 cm.



(a) Calculate the length of *CE*.

Scale factor $= \frac{8}{4} = 2$

$8 \times 2 = 16$

16 cm

(2)

(b) Calculate the length of *BC*.

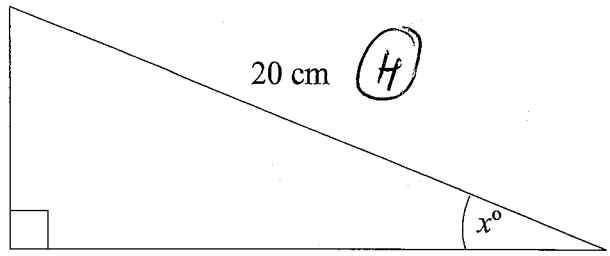
$14 \div 2 = 7$

7 cm

(2)

(Total for Question 8 is 4 marks)

9



Work out the value of x.

$\cos x = \frac{A}{H}$

$\cos x = \frac{13}{20}$

$x = \cos^{-1}\left(\frac{13}{20}\right)$

$= 49.5^\circ$ (1dp)

49.5°

(Total for Question 9 is 2 marks)

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10 Solve the simultaneous equations

$3x + 2y = 9$ $\times 2$
$5x + 4y = 14$

$6x + 4y = 18$
$5x + 4y = 14$

$x = 4$

$3(4) + 2y = 9$

$12 + 2y = 9$

$2y = -3$

$y = -\frac{3}{2}$

$x =$ 4

$y =$ $-3/2$

(Total for Question 10 is 3 marks)